

TE Curve Verification Pipeline Familywise ONNX Report - rcim_track1

TE Curve Verification Familywise ONNX Diagnostics

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates `rcim_track1` paper-reference model-bank archives. Each surface loads the selected harmonic amplitude/phase ONNX components from `models/`, reconstructs full TE curves, and compares those curves against dataset-matched held-out measured TE traces.

Unlike standard familywise model-development exports, `rcim_track1` uses a component bank rather than one ONNX file per surface. The surface tables therefore list archive roots and inventory paths; every exact component ONNX and Python path is recorded in the component inventory CSV. The surface path table intentionally uses archive-root glob patterns because one `rcim_track1` surface is assembled from 19 component ONNX models rather than from a single surface-level ONNX file. The harmonic reconstruction applies the paper-faithful `h0` sign convention per curve direction: forward curves use `-1`, backward curves use `+1`.

Output Artifacts

- output directory: `output/validation_checks/track2_familywise_onnx_report/rcim_track1/2026-07-20-12-53-15__track2_rcim_track1_familywise_onnx_report` ;
- summary YAML: `output/validation_checks/track2_familywise_onnx_report/rcim_track1/2026-07-20-12-53-15__track2_rcim_track1_familywise_onnx_report/track2_familywise_onnx_report_summary.yaml` ;
- model inventory CSV: `output/validation_checks/track2_familywise_onnx_report/rcim_track1/2026-07-20-12-53-15__track2_rcim_track1_familywise_onnx_report/model_inventory.csv` ;
- component model inventory CSV:
`output/validation_checks/track2_familywise_onnx_report/rcim_track1/2026-07-20-12-53-15__track2_rcim_track1_familywise_onnx_report/component_model_inventory.csv` ;
- per-curve metrics CSV: `output/validation_checks/track2_familywise_onnx_report/rcim_track1/2026-07-20-12-53-15__track2_rcim_track1_familywise_onnx_report/per_curve_metrics.csv` .

Simplified Dataset + Setpoints

- dataset: `simplified_dataset` ;
- input mode: `setpoints` ;
- evaluated family: `rcim_track1` ;
- dataset root: `data/simplified_dataset` .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	<code>rcim_track1_simplified_setpoints_forward</code>	<code>historical_simplified_rcim_track1_component_winner_bank</code>	<code>simplified_curve_v1</code>
backward	<code>rcim_track1_simplified_setpoints_backward</code>	<code>historical_simplified_rcim_track1_component_winner_bank</code>	<code>simplified_curve_v1</code>

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	<code>models/simplified_dataset/paper_reference/rcim_track1/forward/*/onnx/*/*.onnx</code>	<code>models/simplified_dataset/paper_reference/rcim_track1/forward/*/python/*/*.pkl</code>
backward	<code>models/simplified_dataset/paper_reference/rcim_track1/backward/*/onnx/*/*.onnx</code>	<code>models/simplified_dataset/paper_reference/rcim_track1/backward/*/python/*/*.pkl</code>
global	<code>not available in archive</code>	<code>not available in archive</code>

Aggregate Metrics

Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.003223	0.003465	7.099	15.049
backward	97	0.004107	0.004324	9.385	22.179

Offset And Shape Metrics

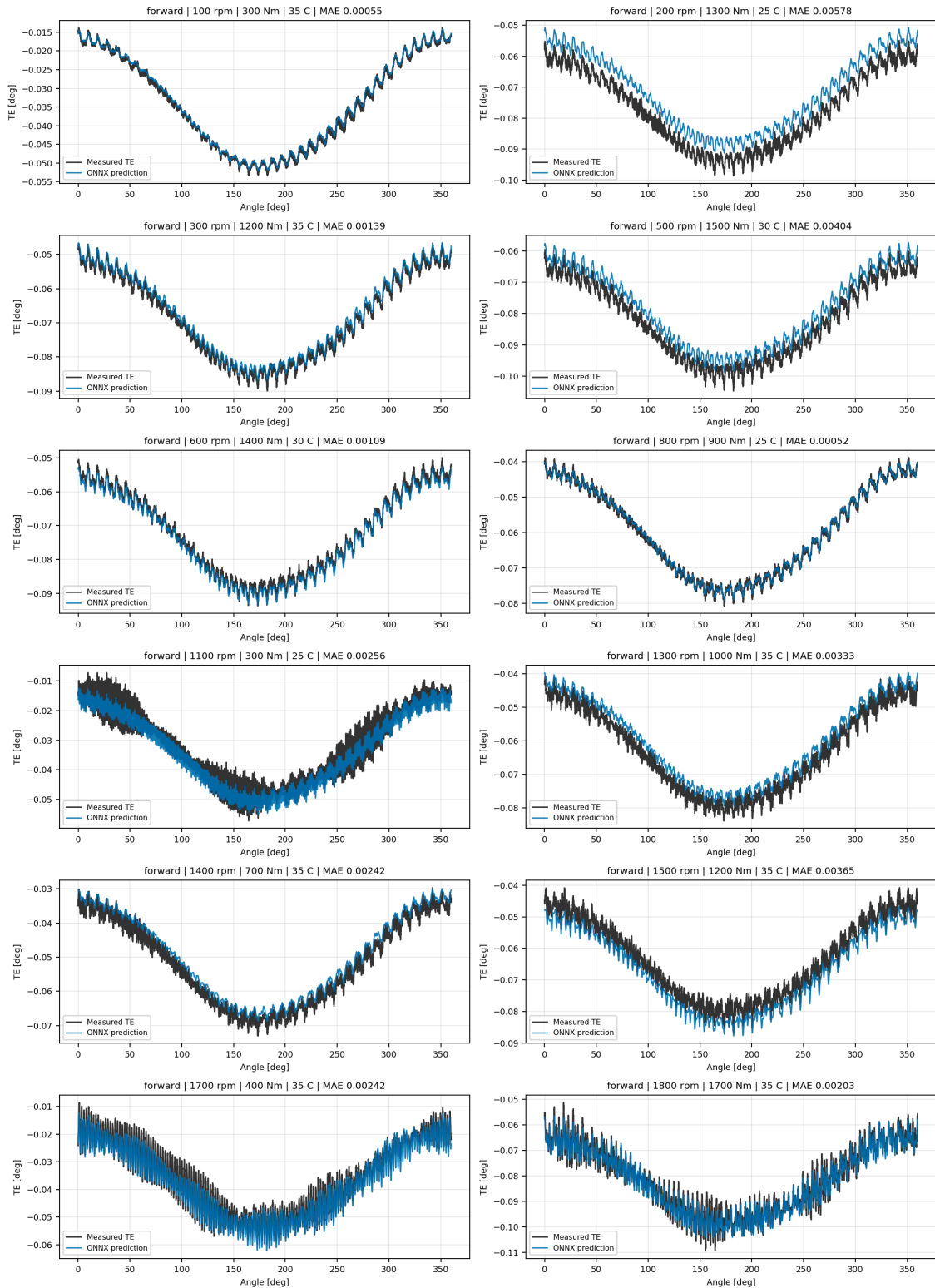
Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000370	0.003028	0.001044	0.004022
backward	0.001922	0.003809	0.001066	0.003162

Unavailable surfaces

- `global` : no archive exists under the dataset-matched `paper_reference/rcim_track1` root.

Forward 12-Curve Page

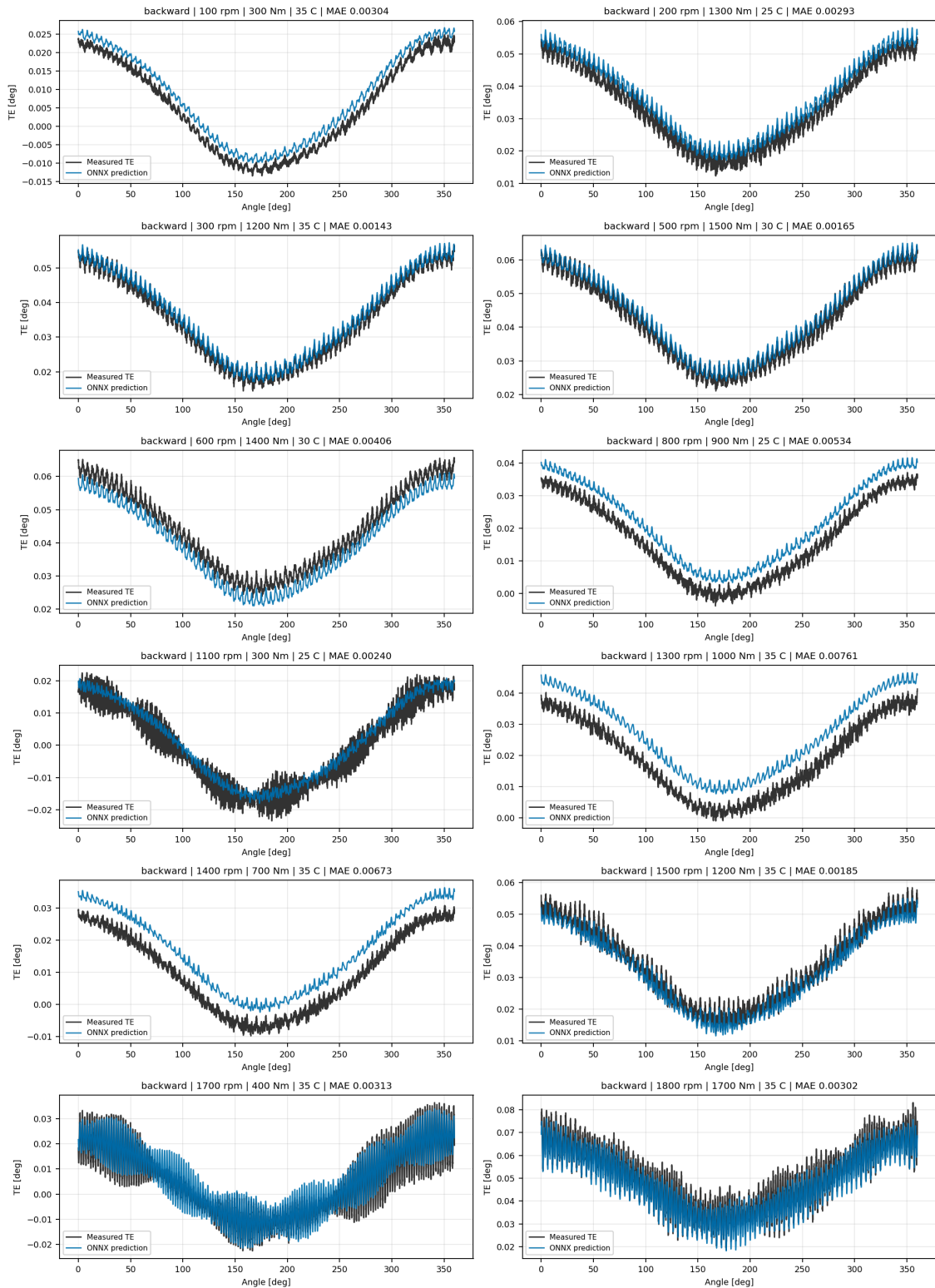
Simplified Dataset + Setpoints | rcim_track1 | forward



simplified_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

Simplified Dataset + Setpoints | rcim_track1 | backward



simplified_dataset_setpoints backward 12-curve collage

Polished Dataset + Setpoints

- dataset: polished_dataset ;
- input mode: setpoints ;
- evaluated family: rcim_track1 ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	rcim_track1_polished_setpoints_fw	2026-07-13-21-11-39_rcim_track1_polished_setpoints_fw_rcim_track1_polished_input_mode_campaign_validation	polished_setpoint_curve_v1
backward	rcim_track1_polished_setpoints_bw	2026-07-13-21-11-39_rcim_track1_polished_setpoints_bw_rcim_track1_polished_input_mode_campaign_validation	polished_setpoint_curve_v1
global	rcim_track1_polished_setpoints_global	2026-07-13-21-11-39_rcim_track1_polished_setpoints_global_rcim_track1_polished_input_mode_campaign_validation	polished_setpoint_curve_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/paper_reference/rcim_track1/setpoints/forward/*/onnx/*/*.onnx	models/polished_dataset/paper_reference/rcim_track1/setpoints/forward/*/python/*/*.pk1
backward	models/polished_dataset/paper_reference/rcim_track1/setpoints/backward/*/onnx/*/*.onnx	models/polished_dataset/paper_reference/rcim_track1/setpoints/backward/*/python/*/*.pk1
global	models/polished_dataset/paper_reference/rcim_track1/setpoints/global/*/onnx/*/*.onnx	models/polished_dataset/paper_reference/rcim_track1/setpoints/global/*/python/*/*.pk1

Aggregate Metrics

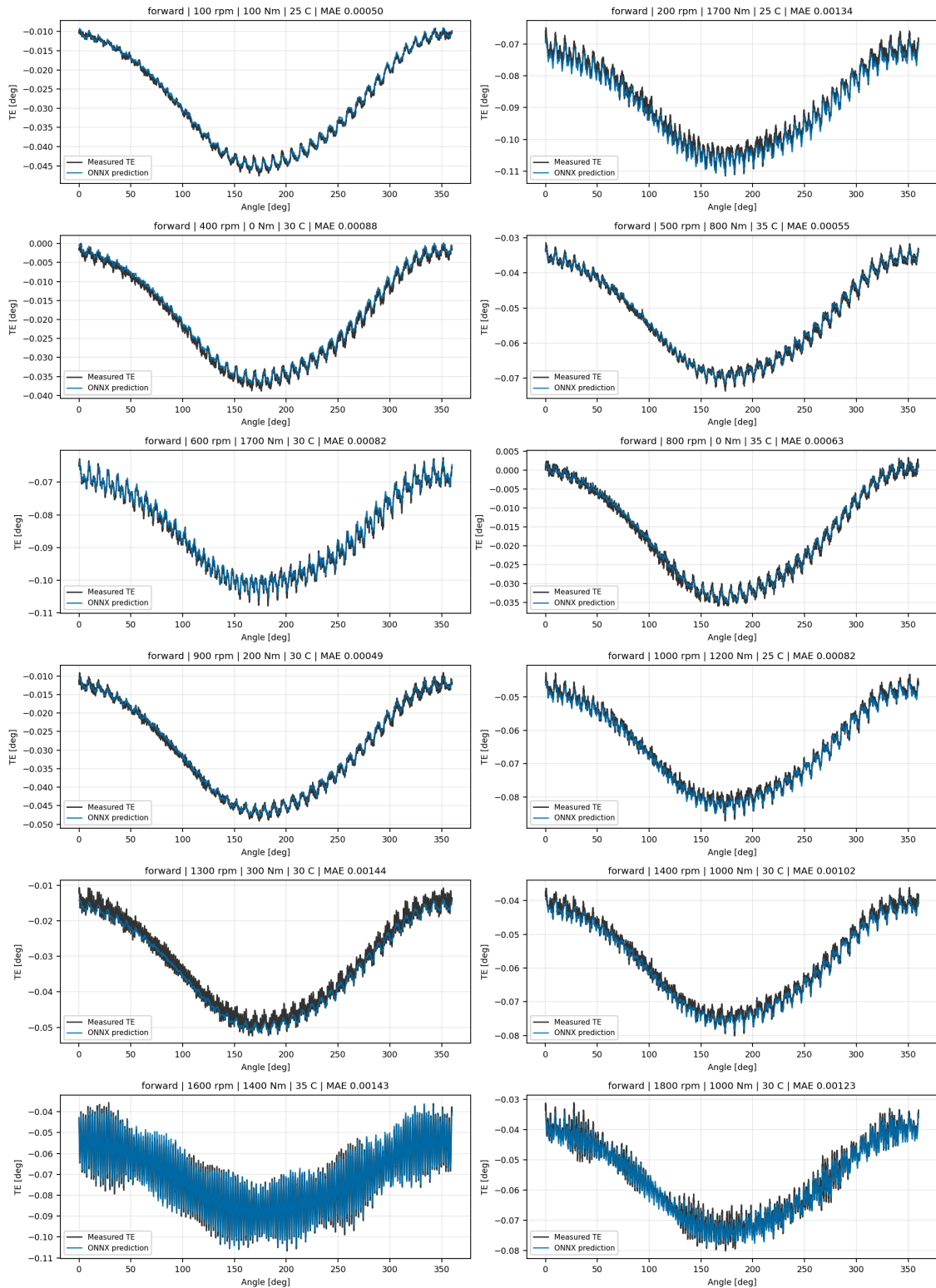
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.001169	0.001389	2.527	5.814
backward	97	0.003582	0.003794	8.356	42.764
global	194	0.002490	0.002756	5.218	21.742

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000111	0.000769	0.000850	0.003634
backward	0.002612	0.003175	0.000970	0.002682
global	0.001207	0.001837	0.001174	0.004152

Forward 12-Curve Page

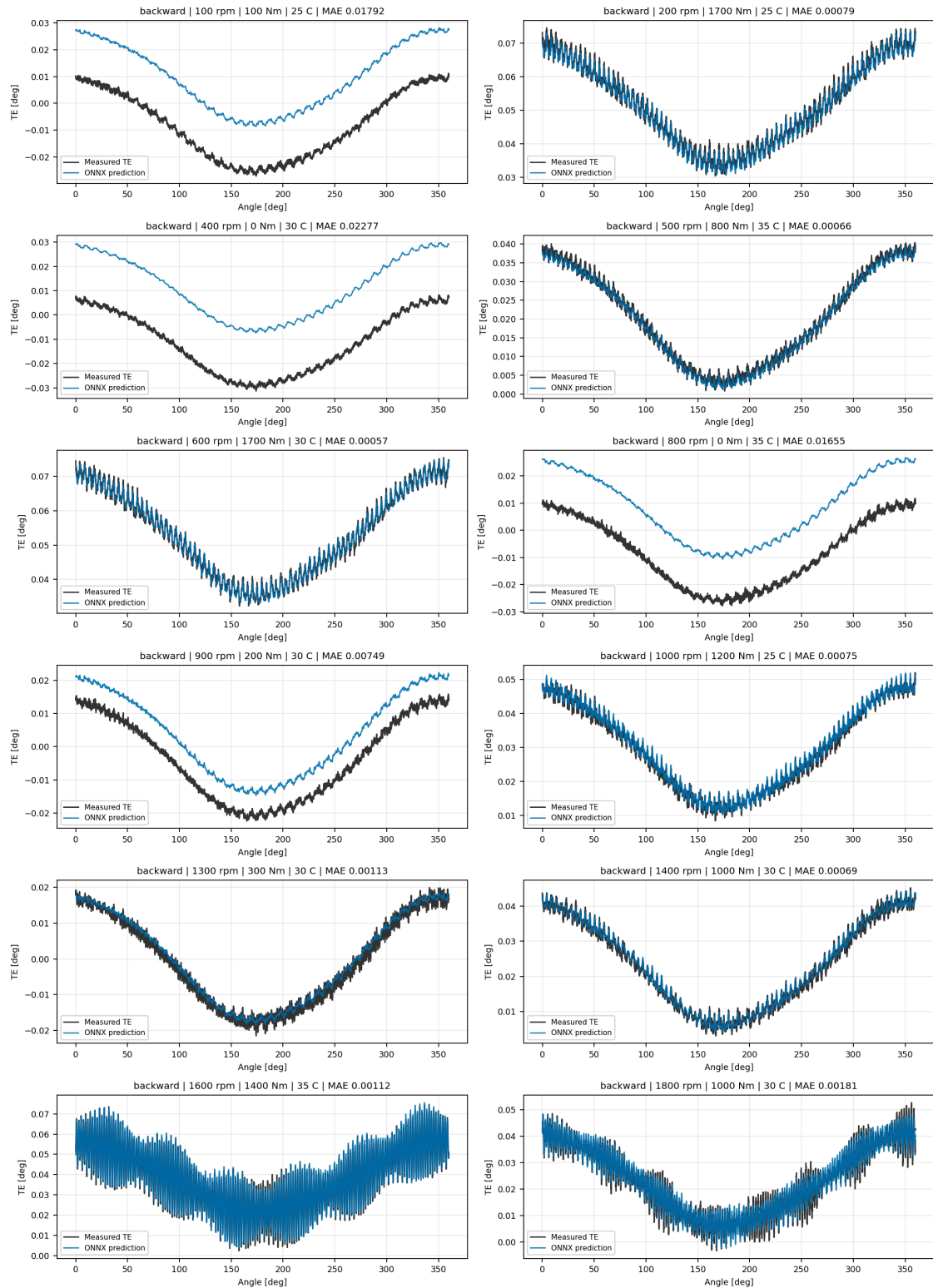
Polished Dataset + Setpoints | rcim_track1 | forward



polished_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

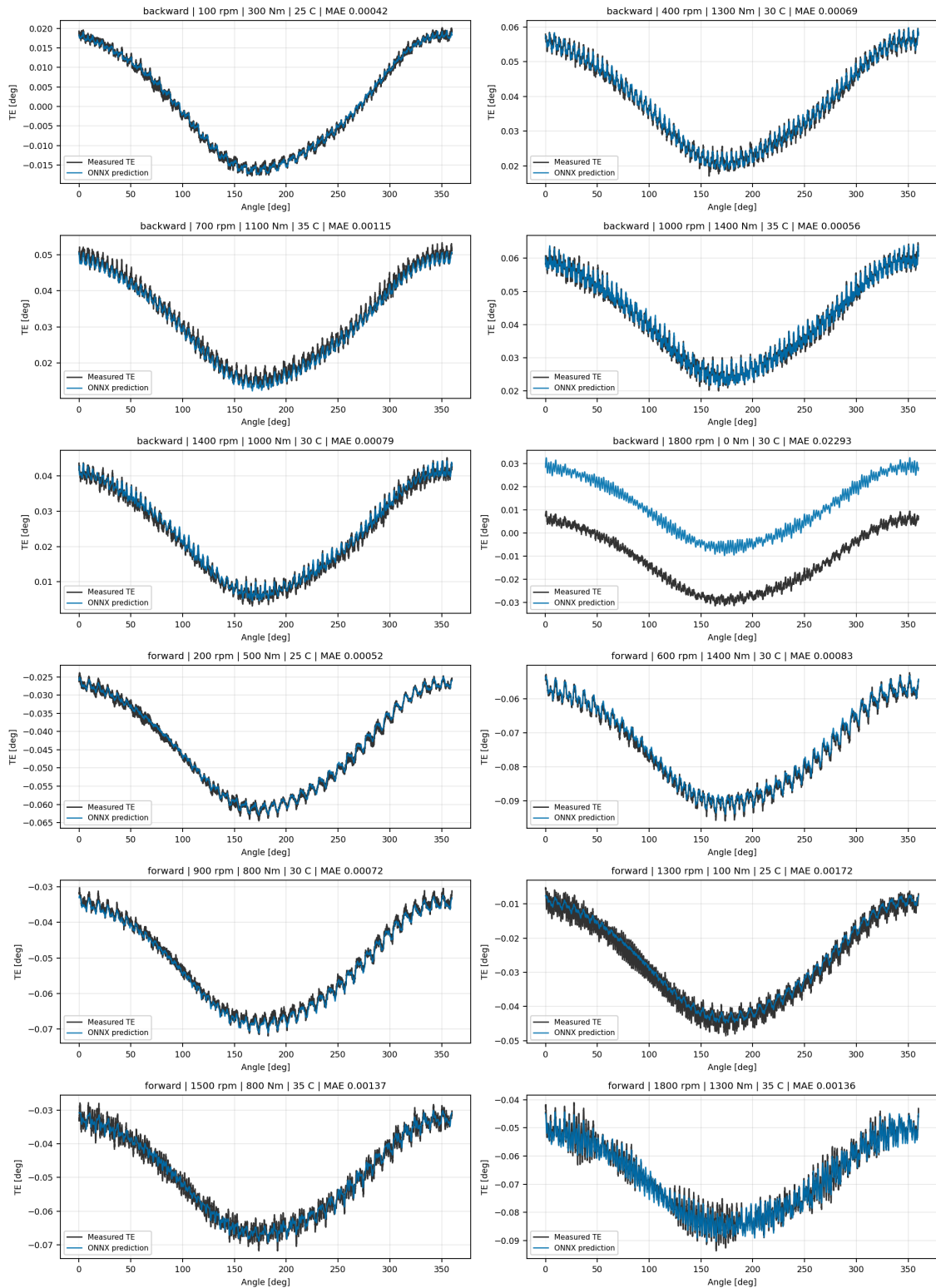
Polished Dataset + Setpoints | rcim_track1 | backward



polished_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Setpoints | rcim_track1 | global



polished_dataset_setpoints global 12-curve collage

Polished Dataset + Actual Values

- dataset: polished_dataset ;
- input mode: actual_values ;
- evaluated family: rcim_track1 ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	rcim_track1_polished_actual_values_fw	2026-07-18-11-55-05_rcim_track1_polished_actual_values_fw_rcim_track1_polished_input_mode_campaign_validation	polished_point_v1
backward	rcim_track1_polished_actual_values_bw	2026-07-18-11-56-46_rcim_track1_polished_actual_values_bw_rcim_track1_polished_input_mode_campaign_validation	polished_point_v1
global	rcim_track1_polished_actual_values_global	2026-07-18-11-55-05_rcim_track1_polished_actual_values_global_rcim_track1_polished_input_mode_campaign_validation	polished_point_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/paper_reference/rcim_track1/actual_values/forward/*/onnx/*.onnx	models/polished_dataset/paper_reference/rcim_track1/actual_values/forward/*/python/*.py
backward	models/polished_dataset/paper_reference/rcim_track1/actual_values/backward/*/onnx/*.onnx	models/polished_dataset/paper_reference/rcim_track1/actual_values/backward/*/python/*.py
global	models/polished_dataset/paper_reference/rcim_track1/actual_values/global/*/onnx/*.onnx	models/polished_dataset/paper_reference/rcim_track1/actual_values/global/*/python/*.py

Aggregate Metrics

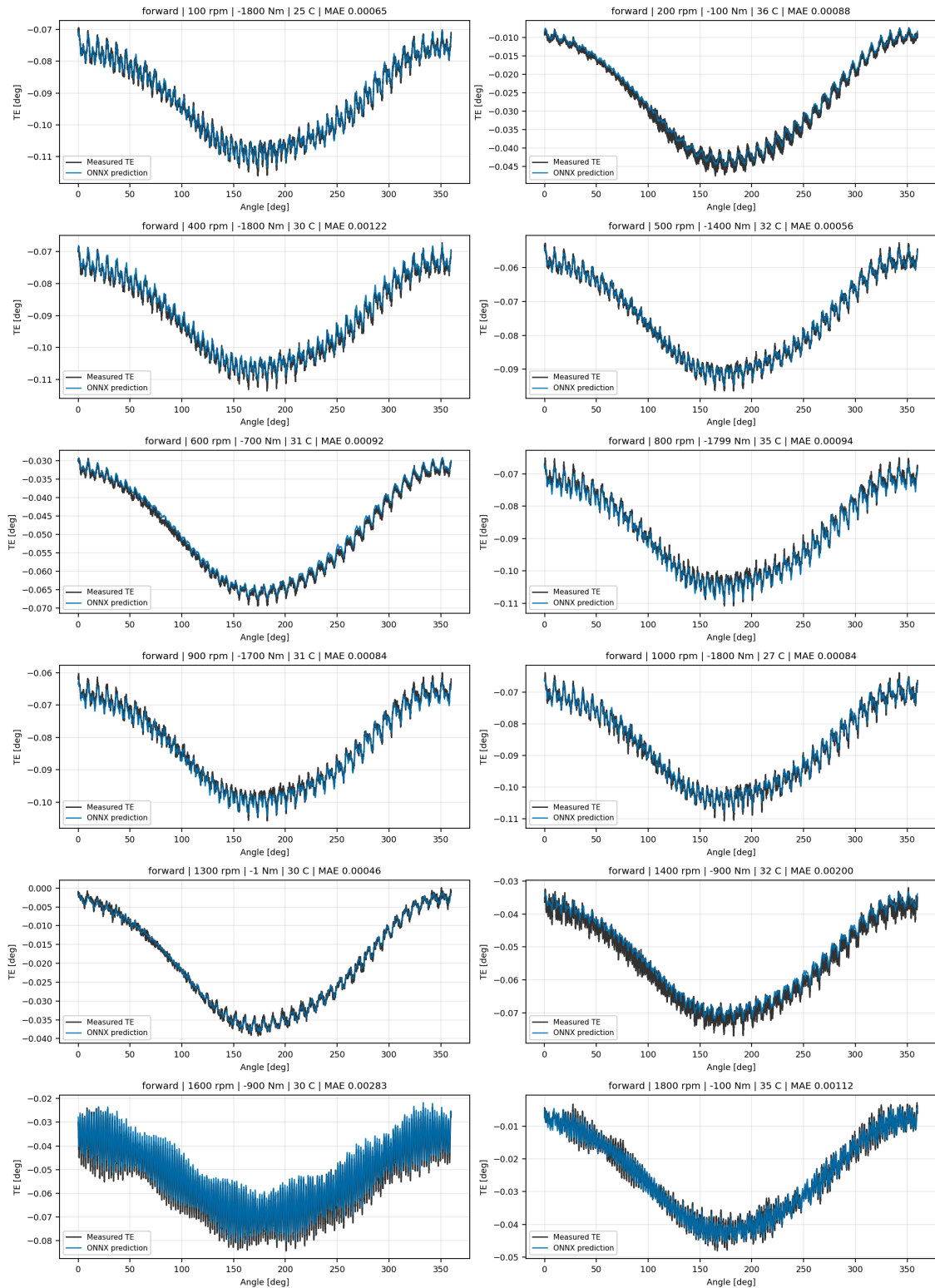
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.001250	0.001469	2.700	6.166
backward	97	0.004765	0.004951	11.063	42.080
global	194	0.002532	0.002810	5.291	20.392

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000039	0.000808	0.000872	0.003662
backward	0.003020	0.004626	0.000920	0.002679
global	0.001284	0.001857	0.001231	0.003953

Forward 12-Curve Page

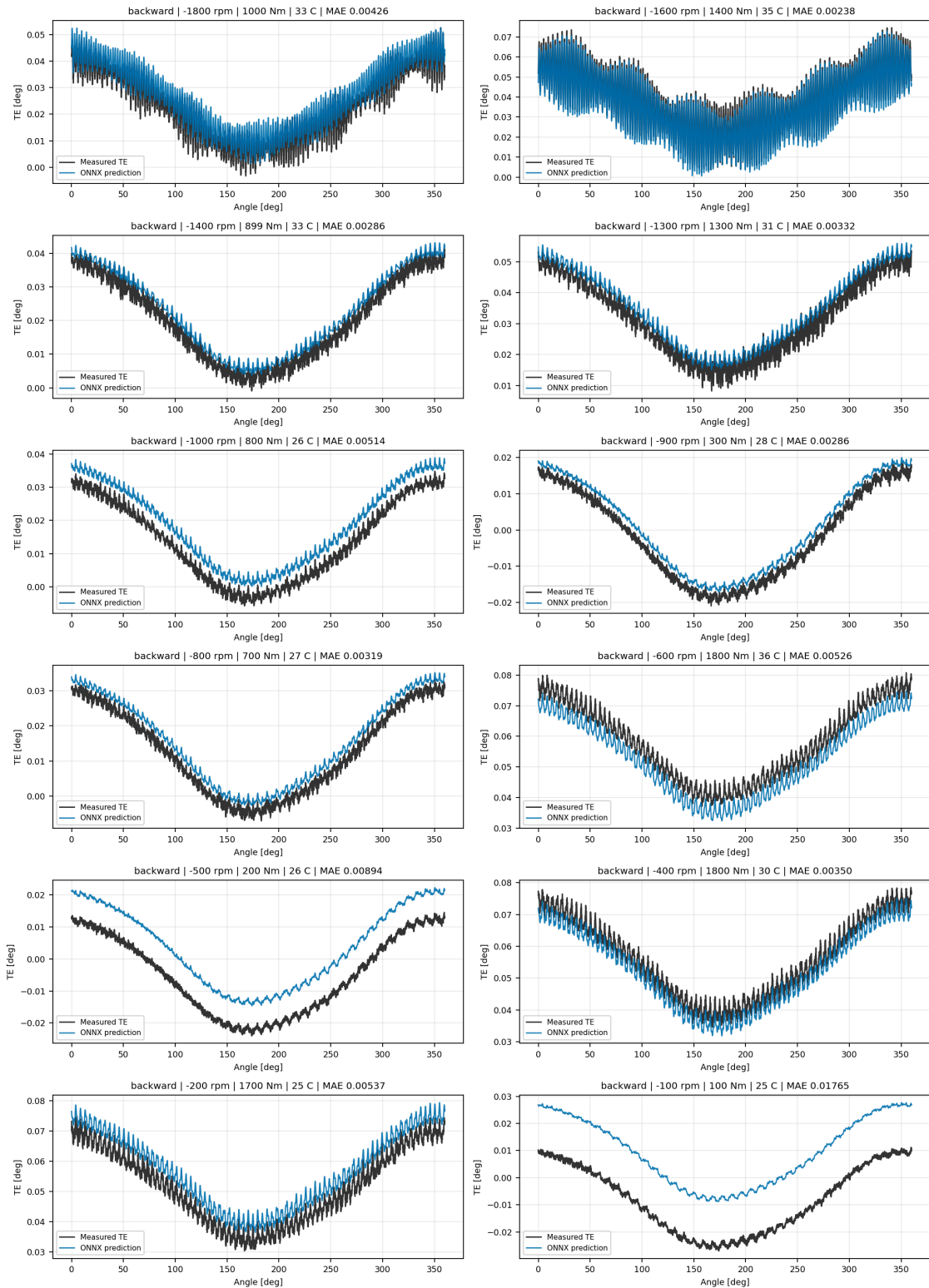
Polished Dataset + Actual Values | rcim_track1 | forward



polished_dataset_actual_values forward 12-curve collage

Backward 12-Curve Page

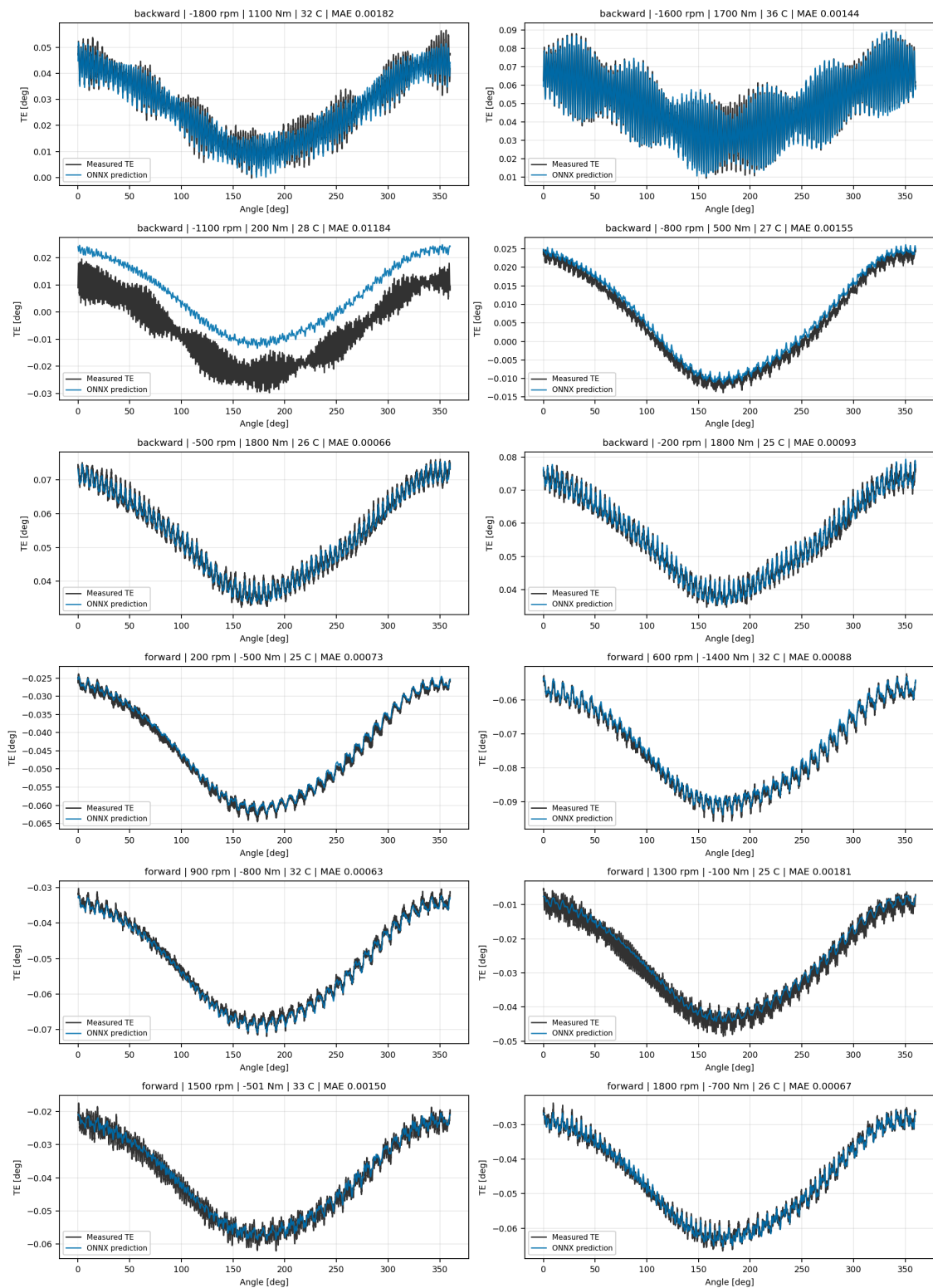
Polished Dataset + Actual Values | rcim_track1 | backward



polished_dataset_actual_values backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Actual Values | rcim_track1 | global



polished_dataset_actual_values global 12-curve collage